BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 064 · Strings and Input Checking

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the result of concatenating 'AB' and '12'?

Q2. How is an item accessed in a 1D array?
A. Using its font
B. Using a file extension
C. Using a network address
D. Using an index

Q3. “Testing a few inputs proves a program can never contain any fault.” Correct this. Explain why.

Q4. A programmer tests input code while building a booking program, then tests the finished program. Correct both claims. Explain why. (a) Iterative testing starts only after delivery. (b) Final testing removes the need for earlier tests.

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I Do · Read new knowledge

Key vocabulary

Length check — A string's .length property can check that a minimum number of characters were supplied.

Validation — Checking entered data against rules improves reliability but does not verify identity.

Input checking — Reject a value that fails a stated rule and request another valid one.

Authentication — Checking a user's identity is different from checking whether input fits a format.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

New learning

Length check: A string's .length property can check that a minimum number of characters were supplied. Apply this in a school portal username.

Validation: Checking entered data against rules improves reliability but does not verify identity. Apply this in a library account.

Input checking: Reject a value that fails a stated rule and request another valid one. Apply this in a club membership.

Authentication: Checking a user's identity is different from checking whether input fits a format. Apply this in a mobile app sign-up.

Worked example

A school portal username is being developed. Explain how length check and validation are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

3. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

4. Two linked points for each concept.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A library account uses length check. Explain how it helps. [2 marks]

Worked method: Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

Guided 2

Explain validation in a club membership. [2 marks]

Worked method: Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

Guided 3

A mobile app sign-up uses input checking and authentication. Explain both. [4 marks]

Worked method: Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Replace the fixed password with an input instruction.

```
password = "abc123"
if password.length >= 6 then
    print("Length OK")
else
    print("Too short")
endif
```

write

Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

Explain how validation and input checking help with a video service username. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A programmer designs a sports team password. Explain why length check, input checking and authentication matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A string's .length property can check that a minimum number of characters were supplied.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. AB12, with no automatically inserted space.

Q2. D — Using an index

Q3. To identify faults and check behaviour against requirements using expected results.

Q4. (a) Iterative testing checks and fixes parts while building the program. (b) Final testing checks the finished program against its requirements. Both are needed.

I Do / We Do pages · Guided practice

Q1. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

Q2. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

Q3. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1].

You Do pages · Code First

```
password = input("Test password:")
if password.length >= 6 then
    print("Length OK")
else
    print("Too short")
endif
```

Expected result: Length OK (if student enters abc123)

You Do pages · Independent questions

Q1. Length check [1]; Input checking [1].

Q2. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: